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819. Two Cases of the Quadric Transformation between Two Planes (concl.)

[p. 101] Writing for convenience (a, b, c) for the coordinates of the fixed point, and (x_1, y_1, z_1) , (x_2, y_2, z_2) for those of the other two points, the formulæ with A, B, C give thus the correspondence

$$x_2 : y_2 : z_2 = bcx_1^2 - a^2y_1z_1 : cay_1^2 - b^2z_1x_1 : abz_1^2 - c^2x_1y_1,$$

which is the first of the two cases in question. These equations give reciprocally

$$x_1 : y_1 : z_1 = bcx_2^2 - a^2y_2z_2 : cay_2^2 - b^2z_2x_2 : abz_2^2 - c^2x_2y_2,$$

or the correspondence is a $(1, 1)$ quadric correspondence.

The formulæ with P, Q, R give in like manner

$$x_2 : y_2 : z_2 = a(ax_1^2 + hy_1^2 + gz_1^2) + (a'y_1z_1 + a''z_1x_1 + a'''x_1y_1), \text{ \&c.},$$

or for shortness

$$x_2 : y_2 : z_2 = a\Omega_1 + a\Omega_2, \quad y_2 = b\Omega_1 + b\Omega_2, \quad z_2 = c\Omega_1 + c\Omega_2,$$

then

$$x_1 : y_1 : z_1 = a\Omega'_1 + a\Omega'_2, \quad \text{\&c.},$$

which is the second of the two cases. We have reciprocally

$$x_1 : y_1 : z_1 = a\Omega_{1'} + a\Omega_{2'}, \quad y_1 = b\Omega_{1'} + b\Omega_{2'}, \quad z_1 = c\Omega_{1'} + c\Omega_{2'},$$

where

$$\Omega_1 = aax_1^2 + by_1^2 + cz_1^2, \quad \Omega_2 = a'y_1z_1 + b'z_1x_1 + c'x_1y_1,$$

and the correspondence is thus in this case also a $(1, 1)$ quadric correspondence.

820. On a Problem of Analytical Geometry

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 203.]

[p. 102] The object of the present note is only to call attention to a problem of Analytical Geometry which presents itself in connexion with the reduction of an algebraical integral, and which is solved, pp. 21, 22 of Clebsch and Gordan's *Theorie der Abel'schen Functionen* (Leipzig, 1866); viz. the problem is considering a line drawn through any given point of a curve $f = 0$ of the order n , to find the value u of α such that which the remaining $n - 2$ points of intersection of the line with the curve f , through the intersection of $\alpha = 0$ are required for the determination of the curve. It is to be shown that we have indirectly an expression for α , and to render the line, in the quadric curve which passes through the remaining two intersections of the line with Ω , and through three given points. Take (ξ, η, ζ) , (ξ_1, η_1, ζ_1) the coordinates of the two given points on the curve f (f is, say of the order $n = 4$); $\theta(x, y, z) = 0$ the equation of the required curve. By the equation of suppositions originally of the form, that the line $x = x' + \rho x''$, &c. joining (x', y', z') , (x'', y'', z'') meets the curve θ where $\theta = 0$ in two points Λ , which when this factor is thrown out becomes

$$a\Lambda^2 + 2b\rho\Lambda + c\rho^2 = 0,$$

where

$$\begin{aligned} a &= (\xi_1, \eta_1, \zeta_1)^2 \theta, & b &= (\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta) \theta, & c &= (\xi, \eta, \zeta)^2 \theta, \\ \gamma &= (\xi, \eta, \zeta)^2 \theta, & \beta &= (\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta) \theta, & \alpha &= (\xi_1, \eta_1, \zeta_1)^2 \theta, \end{aligned}$$

where for shortness f, f_1 are written to denote $f(\xi, \eta, \zeta), f(\xi_1, \eta_1, \zeta_1)$ respectively.

[p. 103] The condition as to the two points obviously is that, making the same substitution $x, y, z = b\xi + a\xi_1, b\eta + a\eta_1, b\zeta + a\zeta_1$ in the equation $\Omega = (a, \dots, b)(x, y, z)^2 = 0$, we must obtain the same quadric equation in $a : b$, the equation in $(a, b) = 0$. We have therefore three conditions, which, introducing an indeterminate multiplier θ , are expressed by the three equations

$$\begin{aligned} (\alpha, \dots, \xi) \xi_1, \eta_1, \zeta_1)^2 &= \theta a, \\ (\alpha, \dots, \xi) \xi_1 \xi, \eta_1 \eta, \zeta_1 \zeta) &= \theta b, \\ (\alpha, \dots, \xi) \xi, \eta, \zeta)^2 &= \theta c. \end{aligned}$$

The conditions as to the three points are obviously

$$\begin{aligned} (\alpha, \dots, \xi) (x_1, y_1, z_1)^2 &= 0, \\ (\alpha, \dots, \xi) (x_2, y_2, z_2)^2 &= 0, \\ (\alpha, \dots, \xi) (x_3, y_3, z_3)^2 &= 0, \end{aligned}$$

and these equations determine the ratios of a, b, c, f, g, h . But to complete the solution the convenient course is to regard the function $\Omega, = (a, \dots)(x, y, z)^2$ as a quantity to be determined, and consequently to join to the foregoing the equation

$$(\alpha, \dots)(x, y, z)^2 = \Omega;$$

we have then seven equations from which (a, b, c, f, g, h) may be eliminated, the result being expressed by means of a determinant of the seventh order

$$|ext[determinant]| = 0,$$

viz. this is an equation of the form $A\Omega = \theta V$, where A is a constant determinant of the sixth order (i.e. a determinant not involving x, y, z) V a determinant of the seventh order, a quadric function of (x, y, z) , obtained from the foregoing determinant by writing therein $\Omega = 0$ and $\theta = 1$; the multiplier θ is and remains arbitrary; but it is convenient to take it to be $= 1$, viz. we then say only that the equation $\Omega = 0$, of the required conic, but we put a determinate value for Ω , say $\Omega = A^{-1}V$, which is, in fact, an equation $\Omega = 0, = A^{-1}V$ arbitrary.

821. On the Geometrical Representation of an Equation between Two Variables

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 210.]

[p. 104] An equation between two variables cannot be represented in a satisfactory manner by a curve, for this serves only to represent the corresponding real values of the variables; to represent the imaginary values the natural course is to represent each variable by a point in a plane; viz. the variable $x, = x + iy$, will be represented by a point the coordinates of which are the components x and y of the variable, and similarly the variable $x', = x' + iy'$, by a point

the coordinates of which are the components x' and y' of the variable; the equation between the two variables will then establish a correspondence between the two points respectively, and to a point on the plane of x there correspond two or more points on the plane of x' . But to complete the solution the convenient course is to regard x, x' as fixed, and respectively, the planes which contain the two points respectively; and it is this correspondence of the two planes which is the proper geometrical representation of the equation between the two variables: to exhibit this correspondence we may in either of the planes draw a network of curves or planes, and then draw in the other plane the network of corresponding curves. This well-known theory [see &c.] illustrated by the notion of a Riemann's surface; taking in the infinite half-plane y positive about the origin as centre a system of semi-circles, to these correspond in the infinite plane of $x'y'$ a series of lunate-shaped curves; and by means of these it is easy to show in the second plane the path corresponding to a given path of the point $x, = x + iy$, in the first half-plane.

822. On Associative Imaginaries

[From the *Johns Hopkins University Circulars*, No. 15 (1882), pp. 211, 212.]

[p. 105] The imaginary (x, y) defined by the equations

$$\begin{aligned}x^2 &= ax + by, \\xy &= cx + dy, \\yx &= \alpha x + \beta y, \\y^2 &= px + qy,\end{aligned}$$

will not be in general associative; to make them so, we must have 8 double relations corresponding to the combinations $x^2.x, xy.x, yx.x, xy.y, y^2.x$ respectively; viz. the first of these gives $x.x.x = x.x.x$, that is, $\theta = x(ax + by) - (ax + by).x, = b(xy - yx), = b(c - \alpha)x + b(d - \beta)y$, and similarly for each of the other terms. We thus obtain by way of the 8 equations between the 6 coefficients $a, b, c, d, \alpha, \beta, p, q$, viz. the relations so obtained are

$$\begin{aligned}b(c - \alpha) &= 0 \text{ (twice)}, & b(d - \beta) &= 0, & p(d - \beta) &= 0 \text{ (twice)}, \\bp - ad &= 0 \text{ (twice)}, & bp - q^2 &= 0 \text{ (twice)}, \\c^2 + dp - c\alpha - p\beta &= 0, & d^2 + bc - bd - dq + f\beta + q\alpha &= 0, \\c^2 + cq - c\alpha - q\beta &= 0, & b(f - d) - q(\alpha - b) &= 0.\end{aligned}$$

From the first four equations it appears that either $b = 0$ and $p = 0$, or else $c = \alpha$ and $d = \beta$; the latter solution may be set aside for the moment, giving the commutative system

$$\begin{aligned}x^2 &= ax + by, \\xy &= yx = cx + dy, \\y^2 &= px + qy.\end{aligned}$$

[p. 106] In order that this may be associative, we must still have the relations

$$\begin{aligned}bp - cd &= 0, \\c^2 + dp - cq - dk &= 0, \\d^2 + bc - ad - bk &= 0,\end{aligned}$$

which are all three of them satisfied by $p = \frac{cd}{b}$, $k = \frac{d^2+bc-ad}{b}$, viz. we thus have the associative and commutative system

$$\begin{aligned}x^2 &= ax + by, \\xy &= yx = cx + dy, \\y^2 &= \frac{cd}{b}x + \frac{d^2 + bc - ad}{b}y,\end{aligned}$$

I did not perceive how to identify this system with any of the double algebras of B. Peirce's *Linear Associative Algebra*, see pp. 120–122 of the Reprint, *American Journal of Mathematics*, t. iv. (1881); but it has been pointed out to me by Mr C. S. Peirce that my system, in the general case $ad - bc \neq 0$, is reducible to the form (a), see p. 122 (l.c.). The object of the present Note is to exhibit in the simple case of two imaginaries the whole system of relations which must subsist between the coefficients in order that the imaginaries may be associative; that is, the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

823. On the Geometrical Interpretation of Certain Formulæ in Elliptic Functions

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 238.]

[p. 107] I MAYBE given in my *Elliptic Functions* expressions for the sn of $u + \frac{1}{2}K$, $u + \frac{1}{2}iK'$, $u + \frac{1}{2}(K + iK')$; but it is better to consider the sn, cn, dn of these combinations respectively, and to write the formulæ thus

$$\begin{aligned}\operatorname{sn}^2(u + \tfrac{1}{2}K) &= \frac{1 - \operatorname{cn} - (1 + i')\operatorname{sn}}{1 - \operatorname{cn} - (1 - i')\operatorname{sn}}, = \frac{1 - i' - (1 + i')\operatorname{sn} - (1 - i')y}{1 - i' - (1 - i')\operatorname{sn} - (1 + i')y}, \\ \operatorname{sn}^2(u + \tfrac{1}{2}iK') &= \frac{(1 + i)\operatorname{sn} + (1 - i)\operatorname{cn}}{(1 + i)\operatorname{sn} + (1 - i)\operatorname{cn}}, \\ \operatorname{cn}^2(u + \tfrac{1}{2}K + \tfrac{1}{2}iK') &= \frac{-(1 - i)\operatorname{sn} + (1 + i)\operatorname{cn}}{1 + (1 - i)\operatorname{sn} + (1 + i)\operatorname{cn}},\end{aligned}$$

where in the last set of ratios x, y are used to denote $\operatorname{sn}^2 u$ and $\operatorname{cn}^2 u$ respectively; and the three combinations above show that for u are written respectively the values just mentioned. The curve $y^2 = (1 - x)(1 - k^2x)$ may, by an inflexion at infinity on the line $x = 0$, and the three tangents from the inflexion are at the points $x, y = 0, 0, \theta, 0, (\frac{1}{k}, 0)$ respectively; hence these points are syntactic points.

[p. 108] We may from any of these, for instance the point $(0, 0)$, draw four tangents to the curve, $(1 + i)x + (1 - i)y = 0$ to the conic, viz. touching at the points $x, y = 0, 0, (\frac{1}{k}, 0)$ respectively; it can therefore be shown how the system of equations to which an arbitrary constants in the system of conics, by introducing in place of the natural conics a system of conics syntactically corresponded; that, in the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

824. Note on the Formulæ of Trigonometry

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 241.]

[p. 108 cont.] The equations $a = c \cos B + b \cos C$, $b = a \cos C + c \cos A$, $c = b \cos A + a \cos B$, which connect together the sides a, b, c and the angles A, B, C of a plane triangle, may be presented in an algebraical rational form, by introducing in place of the angles A, B, C the functions $\cos A + i \sin A$, $\cos B + i \sin B$, $\cos C + i \sin C$ calling these $\frac{x}{y'}$, $\frac{y'}{z'}$, $\frac{z'}{x'}$ respectively, or, what is the same thing, writing $\Sigma \cos A = \frac{x}{y'} + \frac{x'}{y}$, $\Sigma \sin A = \frac{x}{y'} - \frac{x'}{y}$, then the foregoing equations may be written

$$\begin{aligned} (-\Sigma x + y' + z')(-\Sigma x' + y + z) + i \Sigma y \Sigma y' &= 0, \\ (c \Sigma x' + xz - \Sigma x x') - i \Sigma x (y' z' + y' z) &= 0, \\ (c \Sigma y' + x' y' - \Sigma x x') - i \Sigma y' &= 0; \end{aligned}$$

that is, as a system of bipartite equations in (a, b, c) and cubic in (x, y, z) .

Similarly to Spherical Trigonometry, writing as above for the angles, and for the sides writing in like manner $2 \cos a = \frac{l}{l'} + \frac{l'}{l}$, $2 \cos b = \frac{m}{m'} + \frac{m'}{m}$, $2 \cos c = \frac{n}{n'} + \frac{n'}{n}$, we have a system of bipartite equations separately homogeneous in regard to (x, y, z, x', y', z') and (a, B, b, q, R) respectively.

825. A Memoir on the Abelian and Theta Functions

[Chapters I to III, *American Journal of Mathematics*, t. v. (1882), pp. 137–179;
Chapters IV to VII, *ib.*, t. vii. (1885), pp. 101–167.]

[p. 109] THE present memoir is based upon Clebsch and Gordan's *Theorie der Abel'schen Functionen*, Leipzig, 1866 (here cited as C. and G.); the employment of differential rather than of integral equations is a novelty; but the chief addition to the theory consists in the determination which I have made for the cubic curve, and also (but not as yet in a perfect form) for the quartic curve, of the differential expression $d\Pi_n$ (or as I write it $\bar{d}\Pi$); is the integral of the third kind $\int_{1,2} \bar{d}\Pi_n$ in the dual normal form (collide Nuessellen) for which we have (p. 117) $\int_{1,2} \bar{d}\Pi_n = \int_n \bar{d}\Pi_n$, the limits and parametric points interchangeable. The want of this determination presented itself to me as a barrier in the theory during the course of lectures on the subject which I had the pleasure of giving in the Johns Hopkins University, Baltimore, U.S.A., in the months January to June, 1882, and I succeeded in effecting it for the cubic curve; but it was not until shortly after my return to England that I was able partially to effect the like determination in the first more difficult case of the quartic curve. The memoir contains, with additional developments, a reproduction of the course of lectures just referred to, I have endeavoured to simplify as much as possible the notations and demonstrations of Clebsch and Gordan's admirable treatise; to bring some of the geometrical results into greater prominence, and to illustrate the theory by examples in regard to the cubic, the nodal quartic, and the general quartic curves respectively. The various chapters are: I, Abel's Theorem; II, Proof of Abel's Theorem; III, The Major Function (continued); V, Miscellaneous Investigations; VI, The Nodal Quartic; VII, The Functions T, U, V, Θ . The paragraphs of the whole memoir will be numbered continuously.

Chapter I. Abel's Theorem.

Proposition. Art. Nos. 1 to 5.

[p. 110] 1. Starting from the equation $\phi = (x, \dots, \xi, y)^n = 0$ of the variable curve, we have

$$\Sigma(x, y, 1)^{n-2} du = 0.$$

The Differential Pure and Affected Theorems. Art. Nos. 1 to 5.

1. We have a fixed curve and a variable curve, and the differential pure theorem consists in a set of linear relations between the displacements of the intersections of the two curves; in the affected theorem, a linear function of the displacements is equated to another differential expression. I state the two theorems, giving afterwards the necessary explanations.

The pure theorem is

$$\Sigma(x, y, 1)^{n-2} du = 0.$$

The affected theorem is

$$\Sigma(x, y, 1)^{n-2} du = -\frac{b\phi_n}{\phi_n} \frac{d\phi_n}{\phi_n} \frac{d\phi_n}{\phi_n}.$$

2. We have a fixed curve $f = 0$, or say the curve f , or simply the fixed curve; of the order n , with δ dpps, and therefore of the deficiency $\frac{1}{2}(n-1)(n-2) - \delta$. The expression "the dpp" means always the δ dpp of f .

And we have a variable curve $\phi = 0$, or say the curve ϕ , or simply the variable curve, of the order m , passing through the dps and besides meeting the fixed curve in $mn - 2\delta$ variable points.

Moreover, du is the displacement of the current point θ , coordinates (x, y, z) , on the fixed curve, viz. the equation $f = 0$ being satisfied, we have

$$\frac{df}{dx} dx + \frac{df}{dy} dy + \frac{df}{dz} dz = 0,$$

and we thence have

$$\frac{df}{dx} : \frac{df}{dy} : \frac{df}{dz} = y dx - x dy : z dx - x dz : x dy - y dx,$$

so that we have three equal values each of which is put $= du$, viz. we write

$$\frac{y dz - z dy}{\frac{df}{dx}} = \frac{z dx - x dz}{\frac{df}{dy}} = \frac{x dy - y dx}{\frac{df}{dz}} = du,$$

and so as thus defined is the displacement.

[p. 111] 3. $(x, y, z)^{n-3} = 0$ is the minor curve, viz. the general curve of the order $n-3$, which passes through the dpp's, and the general curve of the order $n-3$, z contains $\frac{1}{2}n(n-3) + 1$ terms (viz. $\frac{1}{2}n(n-3) - 1$ terms, but y thereto for the dpps), respectively; and $\theta(t)$ denotes the determinant

$$\theta(t) = |\text{ext}[\text{determinant}]|,$$

so that $\theta(t) = 0$ is the equation of the line joining the points 1 and 2; viz. this is the line (123) in the great curve of order $n-2$ passing through the points of 1, 2.

[p. 112] 4. It is of course important to show, and it will be shown, that the number of independent displacement-relations given by the theorem is equal to the number of independent integral relations between the variable intersections.

5. Observe that the pure theorem gives all the displacement-relations between the variable intersections; for instance, we may help to see the latter equal to the number of independent linear relations between the displacements $\Sigma(x, y, z)^{n-2} du = 0$ for the variable intersections.

Examples of the Pure Theorem—The Fixed Curve a Cubic. Art. Nos. 6 to 13.

6. The pure theorem is next applicable to the case $n = 2$, the fixed curve a conic; it is here given no displacement-relation, and this is as it should be, for the variable intersections of ϕ with the conic are not so independent.

The next case is $n = 3$, $\delta = 0$, the fixed curve a cubic. For proper simplicity the equation is to be taken in the form $f = ax^3 + by^3 + cz^3 + 6lxyz = 0$.

For the variable curve we may take the equation

$$du = \frac{dy}{f_x} = -\frac{dx}{f_y};$$

the two cases being of course in Cartesian form, may then be written

$$du = \frac{dx}{-f_y} = -\frac{dy}{f_x};$$

the first curve here may be a cubic, let the variable curve $\phi, = 0$, be a cubic of order $m, = 1, 2, 3$, &c. as the case may be. For the simplest theorem, the curve ϕ is a line.

[p. 113] The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 , respectively, this is

$$|ext[determinant]| = 0,$$

or, in the notation already made use for such a determinant, $123 = 0$.

7. The equation $du_1 + du_2 + du_3 = 0$, here $du = \frac{dx}{ay^2 + 2lzx}$, or as a particular interpretation, if there is a peculiar interpretation of the corresponding two as arbitrary constants, thereby the the curve $f = 0$, an arbitrary constant; consider then the cubic equation $f = ax^3 + by^3 + cz^3 + 6lxyz = 0$. The equation $du = 0, \phi = 0$ is here for a particular intersection of f, ϕ , taking the logarithms and differentiating with regard to x_1, y_1 , and the corresponding values of the major function in those forms

$$\frac{dx_1}{ay_1^2 + 2lz_1x_1} + \frac{dx_2}{ay_2^2 + 2lz_2x_2} + \frac{dx_3}{ay_3^2 + 2lz_3x_3} = 0,$$

and as taking the logarithms and differentiating in regard to a , considering a as variable, $du_1 + du_2 + du_3 = 0$, viz.

$$\Sigma \frac{1}{ay^2 + 2lzx} \log \left(\frac{by^3}{az^3 + 6lzy} \right) = 0,$$

and this equation the differential equation

$$\Sigma \frac{dx}{ay^2 + 2lzx} = 0$$

is considered to be respectively, but is two then the equation found.

[p. 114] 8. As a particular illustration, we have $\frac{dx}{ay^2 + 2lzx} = du, = 0$, and the integral relation $123 = 0$. I give two new ways of obtaining this—first by differentiation.

For the seven coefficients a, b, c, f, g, h , &c., and the integral relation $123 = 0$. I give a direct verification.

To find f, g, h , where f, g, h are the integral cubic, $f = 0$. If be any constant, $f = \frac{dx}{ay^2 + 2lzx} = du$, and the equation of ϕ as $h, 0$ of the curve, and it may be regarded as the determination for f ; with the dps and the ϕ other points $a, \dots, h$ on the curve.

This being so, considering as variable the original determination f , we have by differentiation

$$\Sigma \frac{1}{ay^2 + 2lzx} \log \frac{dx}{(ax^3 + by^3 + cz^3 + 6lxyz)} = 0,$$

or what is the same, taking the logarithms and differentiating with regard to ourselves a , the equation $\Sigma du = 0$ is satisfied.

We have shown also that the cubic curve $f = 0$, and the cube derived line $\phi = 0$ are now equivalent to the two cases.

9. The case $n = 3$, $\delta = 1$, the fixed curve a nodal cubic, and the variable curve $\phi = 0$, of order m .

We have then

$$\Sigma (x, y, 1)^{n-3} du = 0,$$

viz. this is

$$\Sigma du = 0,$$

which presents itself in the form

$$\Sigma du = -\frac{d\phi}{\phi}.$$

For the simplest theorem, we may take ϕ to be a conic.

[p. 115] $du_1 + du_2 + du_3 = 0$, and the integral relation $123 = 0$. This last equation is an integral of the differential equation $du_1 + du_2 + du_3 = 0$; besides which it is in a particular integral.

But regard one of the three points, say 3, as the fixed point, that is, the line pass through the fixed point 1 and 2. We write $du_1 = du_2$, and the differential equation contains ϕ , while the constant variable.

Let the variable curve be a conic; say the intersections with the cubic are 1, 2, 3, 4, 5, 6. Any five of these points determine the conic, and besides therefore be one of the displacement-relations

$$\Sigma du = 0,$$

which expresses that, the equation $123456 = 0$, which expresses that the six points are on a conic.

We have thus $123456 = 0$ as a particular integral of

$$du_1 + du_2 + du_3 + du_4 + du_5 + du_6 = 0.$$

If, however, we take 5 a fixed point on the cubic, then we have the same equation as the general integral of $du_1 + du_2 + du_3 + du_4 = 0$.

We have here obviously $\Sigma du = 0$ as a particular integral of the cubic; and the integral relation $123456 = 0$ has the same equation as the general integral.

But taking also 5 a fixed point on the cubic, the equation $123456 = 0$, which expresses that the six points are on a conic, of any equation $123456 = 0$ for a fixed and 5 fixed point on the cubic.

We have thus $123456 = 0$ as a particular integral of

$$du_1 + du_2 + du_3 + du_4 + du_5 + du_6 = 0,$$

of order $f = 0$, is in fact a particular integral, and so if we also fix 5, the cubic, the equation expressed integrals.

[p. 116] determinate point, independent of the particular conic through 1, 2, 3, 4 and 5 used for the construction. We have thus a particular integral of the differential equation $du_1 + du_2 + du_3 + du_4 = 0$. That is, in the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

Examples of the Pure Theorem—Fixed Curve a Cubic. Art. Nos. 13 and 2.

13. The fixed curve is taken to be the cubic curve $\phi = 0$, the variable curve of the order $f = 0$ of order n , etc.

We may write the equation

$$\Sigma \theta(\xi t) du = 0,$$

where the equation $\theta(t) = 0$ is the equation of the line joining the points 1, 2, 4, viz. we have $\Omega(1, 1, 2, 3, 4, 5, c, t)$; if so, taking a, b, c the general points, through which the line passes, we are not on the conic, and consequently to jam in the foregoing the equation

$$\Sigma \theta(\xi t) du = 0,$$

where $\theta(t)$ is in fact $123 = 0$, the equation

$$\Sigma \theta(\xi t) du = -\frac{d\theta(\xi t)}{\theta(\xi t)},$$

which presents itself in the case of the cubic curve.

[p. 117] as there is no linear relation between the displacements $\frac{du_1}{f_1}$ and $\frac{du_2}{f_2}$ in the ν th arbitrary point $\bar{2}$ and on the cubic the equation becomes an identity in regard to the two lines du_1, du_2 , respectively.

$$\frac{du_1}{f_1} = -\frac{du_2}{f_2}.$$

Observe that dy_1, dx_1 being functions of x_1, y_1 , &c., is a particular value of integral form

$$dF(x, y) = \frac{x_1}{(y_1 + py_1 + \dots) dx_1} + \dots = 0,$$

where, considering x_1, y_1 , &c. as functions of u respectively, and it is also independent of the equation

$$\frac{du_1}{f_1} + \frac{du_2}{f_2} = 0,$$

that is, we may assume

$$-x_1 y_2' + x_2 y_1' = \theta(x_1, x_2),$$

the constant of integration being to be determined.

The Variable Intersections of the Two Curves—Number of Independent Integral Relations.
Art. No. 13.

13. Suppose $n = 3, \delta = 0$ ($p = 1$); the fixed curve a cubic; and suppose successively the remaining intersections with the cubic; here it is, &c., according to the variable curve is a line, a conic, &c.

If $m = 1$, then two points are the cubic determines the line, and consequently the remaining intersection with the cubic, hence the curve C , depends.

If $m = 2$, then 5 points on the cubic determine the conic, and the remaining intersection with the cubic, hence the curve C, f^2 , will be a determinate point, etc.

If $m = 3$, then 8 points on the cubic determine the cubic, and consequently the remaining intersection with the cubic, hence the curve C, f^3 , will be a determinate point.

[p. 118] If $m = n - 3$, the variable curve is a generic functions of p, q, r , but the dependence on the parameters considered all of 1, 2, 3, &c., f which is a determinant of the variable curve.

The Variable Intersections of the Two Curves—Number of Independent Displacement Relations given by the Pure Theorem. Art. No. 20.

20. The number of displacement-relations given by the pure theorem is = number of integral relations. If $n - 2$, $r = 1, 2, 3$, then if the variable curve is a line, conic, &c., respectively to the number on the cubic by a quartic, &c., is a particular line, etc., and consequently to the equation of a cubic, etc., and to a quartic, with a single point; this can be verified, and the number of independent displacement-relations.

If $m = n - 3$, then by elimination of the parametric points and the number of intersections, etc., we have the general number of relations, etc., as in the same way for the variable curve.

In the same way for f^p depending on the parameters of ϕ^p , in such cases the number of independent integral relations is = 4.

[p. 119] 21. Suppose $n = 3$, $\delta = 1$, the fixed curve a nodal cubic, and the variable curve $\phi = 0$ of order m .

In the simplest case, $m = 1$, we may suppose ϕ to be a line, and the integral relation $123 = 0$, of the line being meeting the cubic the curve $f = 0$, is two points $123 = 0$, that is, we may consider one of these points fixed, $123 = 0$, of two points are independent.

If $m = 2$, then five points on the conic determine the conic, and therefore the remaining three intersections; the curve $f = 0$, is a particular line, of the line is six being independent.

If $m = 3$, then 8 points on the cubic determine the cubic, and therefore the remaining intersection; the equation $123 = 0$ for a particular line, of two points are independent.

In general, $n = 3$, $\delta = 1$, the dpts are $n - 4$ points fixed in the cubic, and is the case of the nodal cubic for the case of the lines linking the bordered points.

The integral expression for the displacement-relations are, then, with for a fixed curve a nodal cubic, etc.,

$$np^q + \frac{1}{2}(n-1)(n-2) = m,$$

which is therefore the equation to be proved.

The Variable Intersections of the Two Curves—Number of Independent Displacement Relations given by the Pure Theorem. Art. Nos. 20.

20. Starting from the equation $\phi = (a, \dots, \xi, y)^n = 0$ of the variable curve, we have

$$\frac{d\phi}{dx} dx + \frac{d\phi}{dy} dy + \frac{d\phi}{dz} dz = 0,$$

$$\frac{d\phi}{da} da + \dots = 0.$$

[p. 120] where ξ, η, ζ denote the coordinates respectively. Then

$$\xi : \eta : \zeta = y dz - z dy : z dx - x dz : x dy - y dx,$$

or putting $z dy - y dz \equiv \xi(x, y)$, then ξ is the displacement-relations which are obtained, i.e. ξ, η, ζ .

[p. 121] where du is an arbitrary local function, $u = ax + by + cz$; that is, multiplied by a noble function $(x, y, z)^{n-3}$ then a particular integral.

Let f denote an arbitrary linear function, $u = ax + by + c$; let J be a particular linear function of order $n - 3$ in the variables f , and u this equation gives the formula

$$\Sigma(x, y, 1)^{n-3} du = J da + ka,$$

introducing the formula

$$\begin{aligned} du &= \frac{y dz - z dy}{f_x} = \frac{z dx - x dz}{f_y} = \frac{x dy - y dx}{f_z}, \\ \frac{du}{dx} &= \frac{1}{f_x} (y d - z dy) \\ &= \frac{1}{f_x} \left(\frac{1}{f} f_x dy - \frac{1}{f} f_y dx \right), \end{aligned}$$

or observing that u , b , c are the differential coefficients $\frac{du}{dx}$, $\frac{du}{dy}$, the form in $J(f, \phi)$ are the variables

$$J(f, \phi) = |ext[determinant]| = \frac{d(\phi, f)}{d(x, y)},$$

the term in $[]$ in the Jacobian, or functional determinant

$$du = \frac{-b\phi}{J(\phi, f)}.$$

22. When the form $du = \frac{-b\phi}{J(\phi, f)}$ is established, then in any one of the three theorems

$$\begin{aligned} \Sigma(x, y, 1)^{n-3} du &= 0, \\ \Sigma(x, y, 1)^{n-3} \frac{-b\phi}{J(\phi, f)} &= 0, \\ \Sigma(x, y, 1)^{n-3} \frac{-b\phi}{J(\phi, f)} &= -\frac{b\phi}{\phi}. \end{aligned}$$

But further, if in the first equation we write $u = a$, and consider the linear function $J(f, \phi)$ denotes the Jacobian

$$J(f, \phi) = |ext[determinant]| = \frac{d(\phi, f)}{d(x, y)}.$$

In these equations the only differential symbol is the b differing the coefficients u , &c.; this represents the variable curve, and besides u , &c., the parametric and the parameter, which is independent of all the others. In this manner we have the original theorem reproduced; for the further reduction of the affected theorem.

The Coordinates (p, σ, τ). Art. Nos. 24 to 26.

[*p. 122*] 24. The letter p has just been used to denote the determinant 612; there is often convenient to denote points 1, 2, 3 coordinates (x, y, z) , (x_1, y_1, z_1) , (x_t, y_t, z_t) respectively; and then employing p , r to denote the determinants 623, 631, 612 respectively, and Δ the determinant 123, we have identically

$$\Delta x = x_1 p + x_t \sigma + x z', \quad \Delta y = y_1 p + y_t \sigma + y z', \quad \Delta z = z_1 p + z_t \sigma + z z',$$

which equations, regarding therein the point θ , coordinates (x, y, z) , as a current point, are in fact equations for transformation from the coordinates (x, y, z) to the coordinates p, σ, τ belonging to the triangle of reference 123. The points 1 and 2 have therein the coordinates $(\sigma, \tau) = (1, 0)$, $(\sigma, \tau) = (0, 1)$ respectively.

25. If the function f , which equated to zero gives the equation of the curve, is of the order n in (x, y, z) , then it will be of the order n in (ρ, σ, τ) ; the coefficient of $\sigma^n \rho^n$ is $= \frac{1}{\delta^2} (x_1, x_2)^{n-2} f$; that of $\rho^{n-1} \sigma$ is $= \frac{1}{\delta^2} (x_1, x_2)^{n-1} (x_1) f$; and so on.

But further, if f vanishes at the point 1 to order a , and at the point 2 to order β , then for any line $(\rho\sigma\tau)$ such that $\rho = 0$ does not satisfy $f = 0$, the lower coefficients $\sigma^n \rho^{n-a-\beta}$ vanish:

that is, the coefficient of σ^a in $(\sigma\tau)^a$ is $= 0$, $(a+1)$ coefficients vanish; and similarly $\beta+1$ coefficients of those involving ρ to the highest powers $r^{n-1}, r^{n-2}, \dots$ are in the form

$$f = \frac{1}{\delta^2}(\sigma^n \rho^0 + a\rho\sigma^a + \dots).$$

[p. 123] The terms independent of ρ contain, it is clear, the factor ρ^n , and consequently those terms from the others, the equation may be written

$$f = \rho^a \sigma^\beta + (\rho, \sigma, \tau)^{n-2}.$$

26. The equation $f = 0$, $\rho^a \sigma^\beta = 0$, $\rho^{n-a-\beta} = 0$ determines the residues of the points 1, 2, and hence the major function on $f = 0$, $(\rho\sigma\tau)^{n-a-\beta}$ at the points 1, 2, and writing therein $a = 0$, $b = 0$, the corresponding value of the major function in those forms

$$\frac{1}{n} \frac{1}{x_1^n x_2^n} - \frac{1}{x_1^{n-1} x_2 - \dots} - \frac{1}{x_1 x_2^{n-1}}.$$

27. [p. 123 cont.] we have du being the major function on the cubic, when the equation $A\Omega = 0$ for any major function u , by the equation now obtained, is

$$f = \rho^a \sigma^\beta (\rho, \sigma, \tau)^{n-a-\beta}.$$

The Proposition for the Affected Theorem. Art. No. 27.

27. In the affected theorem instead of (x, y, z) we have two coordinates (ρ, σ, τ) . We have thus to show that the differential expression $J(f, \phi)$ depending also on the parameters of ϕ , may, by the addition of the new variable, by some particular formulæ expressed by $A\Omega + B\Omega$, where A and B are constants. The dependence on the parameters of ϕ may also be eliminated, in many cases by the rest of the parametric points which are linear in the differentials of the parameters. It follows that, $\Delta J(f, \phi)$ depends only on the variation of f . We have consequently

$$\frac{1}{J(f, \phi)} = \Sigma A \frac{\partial \theta_1}{\partial x_1} J(\phi, f),$$

and the equation for the affected theorem becomes

$$\Sigma (x, y, z)^{n-3} \frac{b\phi}{J(\phi, f)} = -\Sigma \left(\frac{ba_1}{x_1} + \frac{b\theta_1}{\theta_1} \right),$$

where (σ, y) , σ , b , etc. to be regarded as standing for its ratios in terms of $J(\phi, f)$.

Jacobi's Fraction-Theorem. Art. No. 28 to 31.

28. This is the extension of a well-known theorem, which, in a somewhat disguised form, may be thus written: viz. if T be any rational and integral function $(x, y)^m$ then we have

$$\frac{1}{\theta} = \Sigma \frac{1}{x_1 - x'} J(T').$$

[p. 124] or, introducing an arbitrary constant A by the equation $A\theta = X$, may this is

$$\frac{A}{X} \left(\frac{1}{x} \right) = \Sigma (\sigma, y)^m F'(x),$$

where U' is the same function $(x', y')^m$ of x' , y' that U is of (x, y) , and $J(\phi, f) = \frac{d(U)}{d(x, y)}$, the Jacobian of the same function (x, y) , and the summation extends to all roots x' of the equation $U' = 0$ obviously in this gives the theorem the formula for the decomposition of $\frac{1}{X}$ into simple fractions.

29. Take now $U = (x, y, z)^m$, $V = (y, z, \dots)^p$, $V = \dots$, functions of (x, y, z) of the degrees m and n , respectively; and then writing as written above the system $U = 0$, $V = 0$; viz. U is any root of $V = 0$, and any other quantity (or it may be) is the summation extends to all the same intersections UV , where the $J(U, V)$ is the Jacobian

$$\frac{1}{AU(x, y, z)} = \Sigma \frac{1}{x - x'} J(U, V).$$

and the equation for the affected theorem becomes

$$\frac{AD - BC}{XY} = \Sigma \frac{1}{x - x' \cdot y - y'} J(U, V),$$

where U' , V' are the same functions of (x', y') that U , V are of (x, y) , and $J(U, V) = \frac{d(U, V)}{d(x, y)}$, the Jacobian; the summation extends to all roots x' , y' of the equations $U' = 0$, $V' = 0$.

[p. 125] 30. For the proof, we write $X = R$, Y a entire function $(x, y)^m$ we may write $AD = -X$, $C = -Y$, and the integral relations $123 = 0$, $X = AY$; the equations imply that the curves $X = 0$, $Y = 0$ meet in m points, all of them on the cubic $f = 0$, but it has been found out to me by Mr C. S. Peirce that we have $(AD - BC)(x, y) = -CX + DY = -AX = 0$, the original relation may be omitted, viz. we have

$$(AD - BC)x = -CX, \quad (AD - BC)y = -DY,$$

and we have given the conditions corresponding to it in the form of pairs of equations, i.e. $(AD - BC)$, $(CU + DV)$, that is, $(AD - BC)U + DY = 0$, where A, B, C, D are arbitrary constants. It follows that any other formula existing, the coefficients of $\sigma^n \rho^n$ as we may write

$$AD - BC = K(x, y).$$

where the summation extends in all the conditions of the use of one for these of x, y , but with a formula relating, the coefficients K may be determined from the conditions, but as is rigid one A, B, D , the multiplication takes place over the formula, and the addition is the some thing,

$$AD - BC = K(x, y), \quad \dots,$$

where the summation extends in all K the combination of the use values of x' and y' at the same root, so it is verified.

10. [p. 126] In all that follows, the cubic might be any cubic whatever, but for greater convenience we suppose it of a fixed form, viz. a binary or cubic; and consider the equation $f = ax^3 + by^3 + cz^3 + 6lxyz = 0$. Such a cubic has the three inflexions $(x, y, z) = (0, 0, 1)$, $(0, 1, 0)$, $(1, 0, 0)$ at the vertices of the triangle of reference and the three inflexional tangents are at the sides of the same triangle; viz. the cubic has the inflexional tangent at the points 1, 2, 3 are sides of the triangle of reference, and the three inflexional tangents are at the sides; the inflexion $(0, 1, 0)$ is the the side, that is, $z = 0$, $cz^3 = 0$; etc., respectively, that is to say the cubic has the three inflexional sides of the reference $(0, 0, 1)$, $(0, 1, 0)$, $(1, 0, 0)$.

Consider, in the cubic, are the coordinates $(\rho, \sigma, \tau) = (0, 0, 1)$. We have

$$\rho^a \sigma^\beta (\rho, \sigma, \tau)^{n-a-\beta} = \rho^a \sigma^\beta \cdot \tau^{n-a-\beta}.$$

11. [p. 127] The left-hand side is

$$A(D - BC) \Sigma \frac{1}{x - x'} \Sigma \frac{1}{x' - x_2} - \Sigma \frac{1}{y - y'} \Sigma \frac{1}{y' - y_2},$$

and $A(D - BC)$ the some of the highest order in (x, y, z) $\phi = 0$, $\phi = 0$ being a homogeneous function of the lowest degrees $n + n' = m$, of the order ϕ may be written

$$\Sigma(AD - BC) = \Sigma(x, y, z)^m F'(x, y, z),$$

and on the linear formula $AX = R + S$; A, B, C, D are entire functions of X, Y in their highest degrees, and a formula relating ϕ also, in particular ϕ , the formula is associated.

There is no thing so, we may consider the formula as well known under the system of conditions of those points are $(AD - BC)$, with the assumption (x, y) , the quotient $(AD - BC)\rho^a\sigma^\beta(\rho, \sigma, \tau)^{n-a-\beta}$, of the quotient $A\rho^a\sigma^\beta(\rho, \sigma, \tau)^{n-a-\beta}$, can in two states again be written

$$b = \text{const. of } (x, y),$$

comparing the two values of y , we have

$$\text{Const. of } \frac{(AD - BC)\rho^a\sigma^\beta(\rho, \sigma, \tau)^{n-a-\beta}}{a(\sigma)^{n-1}Y'} = \Sigma \frac{(\sigma)^{n-1}}{(\sigma, y)^m F'(x)},$$

and we hence derive

$$\text{Const. of } \frac{(AD - BC)\rho^a\sigma^\beta(\rho, \sigma, \tau)^{n-a-\beta}}{a(\sigma)^{n-1}Y'} = \Sigma \frac{(\sigma)^{n-1}}{(\sigma, y)^m F'(x)}.$$

[p. 128] where $(x, y)^{m+n-2}$ is the general function of the degree $m + n - 2$, and, of course, $(x', y')^{m+n-2}$ is the same function of x', y' . The two functions may be written $(x, y, 0)^{m+n-2}$ and $(x', y', 0)^{m+n-2}$; and this being so we may on the right-hand write instead $(x', y', 1)^{m+n-2}$, for, by so doing, we introduce in the numerator of the fraction new terms of an order not exceeding $m + n - 3$, and by the $(m + n - 3)$ theorem already obtained the sum Σ of the quotient of such terms by $J(U', V')$ is $= 0$. We thus have

$$\text{Const. of } \frac{(AD - BC)_0(x, y, 0)^{m+n-2}}{\alpha\beta(xy)^{mn-1}} = \Sigma \frac{(x', y', 1)^{m+n-2}}{J(U', V')}, \quad \text{say the } (m + n - 2) \text{ theorem,}$$

where $(x', y', 1)^{m+n-2}$ is the general non-homogeneous function of the degree $m + n - 2$, and $(x, y, 0)^{m+n-2}$ is obtained from it by attending only to the terms of the highest degree $m + n - 2$, and therein substituting x, y for x', y' .

35. We may, it is clear, in the equations for the $(m + n - 3)$ and for the $(m + n - 2)$ theorems respectively, omit the accents on the right-hand sides; doing this, and moreover in each equation transposing the two sides, the two special theorems are

$$\Sigma \frac{(x, y, 1)^{m+n-3}}{J(U, V)} = 0, \quad (m + n - 3) \text{ theorem,}$$

$$\Sigma \frac{(x, y, 1)^{m+n-2}}{J(U, V)} = \text{const. of } \frac{(AD - BC)_0(x, y, 0)^{m+n-2}}{\alpha\beta(xy)^{mn-1}}, \quad (m + n - 2) \text{ theorem.}$$

Homogeneous Form of the Theorem. Art. Nos. 32.

36. Writing $\frac{x}{z}$ for x , $\frac{y}{z}$ for y , and introducing the suppositions of the arbitrary linear functions $u = ax + by + cz$, we may obtain, viz. a, b, c as variables; we have

$$\begin{aligned} \Sigma \frac{1}{J(U, V)} du &= 0 \quad (n = n - 2) \text{ (theorem),} \\ \Sigma \frac{1}{J(U, V)} du &= -\text{const. of } \frac{(AD - BC)}{J(U, V)} \quad (n = m + n' - 2) \text{ (theorem),} \end{aligned}$$

where the suffix 0 denotes that we are in $AD - BC$ to write $z = 1$.

If in the last formula we change throughout b into $-b$, the result may then be expressed in two terms in b , but a still containing only a , by σa to be put $= a$, by which it appears in this case the formulæ the particular function $(AD - BC)\rho^a\sigma^\beta(\rho, \sigma, \tau)^{n-a-\beta}$.

[p. 129] The effect of $\frac{b\phi}{\phi}$ of the Curves $U = 0, V = 0$. Art. Nos. 37 and 38.

37. We want, in regard to the foregoing special theorems, consider the effect of any change of the values of U, V, f , etc.

Suppose now we have any of the foregoing functions, where U, V are not put now passes through it; the dpts of an intersection of U, V , and the theorems are similarly true.

If U passes through the dpts, then the corresponding formula remains, viz. taking $\frac{1}{V}$ as a constant of the formula. But it must, that, the dps in homogeneous as the values of $U, V, \dots$ by some constant, we have it correspondingly through the theorem on the cubic.

[p. 129 cont.] Suppose now we have any change of the values of V, f , by some pass through it; the dps in any intersection of U, V , and the theorems are similarly true.

The Syzygy. Fixed Curve a Cubic. Art. No. 35.

[p. 130] 38. Writing Ω_3 , as before, we now consider the curve V , has a dp, but taking on the fixed curve a new point, $V = 0$, in fact be deduced from any equation $f = 0$ of the cubic.

Suppose so, and the integral relation $123 = 0$, will be that the equation of the cubic curve, we ought to have $(123)\Omega = 0$, suppose

$$[\theta\Omega] + (\theta\Omega) + (\Omega) = 0,$$

and the form, may, by direct consideration on the left-hand side as expressed in terms of (x, y, z) , viz. if instead of these being satisfied as before, but in particular they give

$$[\theta\Omega] + (\theta\Omega) + (\Omega) = 0;$$

this, multiplied by ρ , becomes

$$\frac{1}{J(\phi, f)} \rho d\sigma + b\rho d\sigma + \sigma b d\sigma = 0.$$

And here, observing 1, V is to be a function of the form $f = 0, V \cdot \Omega = 0$, the equation of three points constants. Observing that the three points 1, 2, 3 are also on the curve $\frac{x'p}{x'^2}$, then the conic possesses the line connecting the points and we obtain on these as similar conditions for $\frac{x'p}{x'^2}$, with which is a feature of ρ , this is, the

$$(VOLY)\rho = -(V\Omega)\sigma - (V\Omega)\rho = 0,$$

or, on substituting for the value of $(VOLY)$, we get

$$\rho d\sigma + \sigma b d\rho + (V\Omega) = 0,$$

which is the required syzygy.

39. [p. 131] But for $\tau = 0$, we have

$$A_1\phi_1 + B_1\phi_2 = a\rho^m, \quad C_1\phi_1 + D_1\phi_2 = b\rho^m,$$

and hence

$$(AD - BC)\phi_1 = A_1\phi_1 B_1\phi_2 - B_1\phi_1 A_1\phi_2 = C\rho^m,$$

and the right-hand side these becomes; say

$$-\theta \text{ const. of } \left(\frac{AD - BC}{f_x f_y} \right) \frac{1}{\rho^a \sigma^b},$$

But in calculating the constant of $\frac{AD-BC}{f_x f_y}$ has but suppose not only $\tau = 0$ but also $\rho = 0$, we then have $f_x f_y = \left(\frac{\tau}{V}\right) (x, y, z)^{n-2}$; that is, $\frac{f_x f_y}{\rho^a \sigma^b} = \left(\frac{\tau}{V}\right) (x, y, z)^{n-2}$ and hence also $\frac{f_x f_y}{\rho^a \sigma^b} = \left(\frac{\tau}{V}\right)$. Similarly, in calculating the constant of $\frac{AD-BC}{\rho^a \sigma^b}$, we may suppose not only $\tau = 0$ but also $\rho = 0$; we then have $f_x f_y = \left(\frac{\tau}{V}\right) (x, y, z)^{n-2}$, and similarly writing in the second equation $\rho = 0$, we find $C_1 \left(\frac{\tau}{V}\right) = D\rho^m$, that is, $\frac{C_1}{\rho^m} = \left(\frac{\tau}{V}\right)$, and the expression then becomes

$$= -\Delta \left(-\frac{b\rho_1}{\rho_1} + \frac{b\rho_2}{\rho_2} \right),$$

giving the equation which was to be proved. This completes the proof of the affected theorem

$$\Sigma \frac{(x, y, z)^{n-3} du}{J(f, \phi)} = -\Sigma \frac{b\rho_1}{\rho_1} - \frac{b\rho_2}{\rho_2}.$$

Chapter III. The Major Function $(x, y, z)^{n-3}$.

Analytical Expression of the Function. Art. Nos. 42 to 48.

[p. 132] 42. The function has been defined by the condition that the curve $(x, y, z)^{n-3} = 0$ shall pass through the dps, and also through the $n - 2$ residues of the parametric points 1, 2; and moreover, that on writing therein (x_1, y_1, z_1) for (x, y, z) , the function shall become $= 1 \div p_1$. Obviously the function is not uniquely determined; calling the curve $\Omega = 0$, we may add to Ω any term $\theta J + \dots = 0$, J a constant factor. We may then take the original curve, we ought to have $\Omega = 0$. We have $\Omega = \rho + (\rho, \sigma, \tau)^{n-3}$. We may a multiple of $\theta J + \dots$.

43. We can without difficulty write down an equation determining Ω' as a function $(x, y, z)^{n-3}$, on putting therein (x_1, y_1, z_1) for (x, y, z) the function becomes $= 1 \div p_1$. In the case where $\rho = \theta + bp$, $\sigma = bp$, $\tau = bp$, the function is a quartic, in the value of the conic function Ω , in the form

$$\Omega = \rho + (\rho, \sigma, \tau)^{n-3},$$

and is in the form $\Omega = \rho + (\rho, \sigma, \tau)^{n-3}$, which may be developed in $(\rho, \sigma, \tau)^{n-3}$ and is in two cases as θ also as a curve and the residues. The equation has the form

$$\Omega = \rho\sigma\tau + (\rho, \sigma, \tau)^{n-3}.$$

But the bipartite curve also $J = 0$, the residue of the parametric points 1 and 2 are then a fixed point. Suppose, then, the function of the lowest degrees $(\sigma\tau)^{n-3}$, the bipartite to in regard to (σ, y) , are equal but obvious, the points and indices of the fixed points; the major function being equal to 0 is in fact the

$$\Omega = \frac{1}{J(\sigma\theta J)^{n-1}}.$$

We have $\Omega \neq 0$, $V = 0$, and the same equation determines for the cubic, the residues, and these for the cubic.

[p. 133] 44. Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic; and no homogeneous as which is to be shown that we have determinations of Ω , viz. at $(\rho\sigma\tau) = (0, 0, 1)$, $\theta J = 0$, the cubic conditions $\theta J = (0, 1, 0)$ to a residue. Equation $\Omega = 0$ at $\rho = 0$, and through the points and through the remaining two intersections of the line with Ω , and through the three given points. Take (ξ, η, ζ) , (ξ_1, η_1, ζ_1) the coordinates of the two given points on the

curve f (f is, say of the order $n = 4$); $\theta(x, y, z) = 0$ the equation of the required curve. By the equation of suppositions originally of the form, that the line $x = x' + \rho x''$, &c. joining (x', y', z') , (x'', y'', z'') meets the curve θ where $\theta = 0$ in two points Λ , which when this factor is thrown out becomes

$$a\Lambda^2 + 2b\rho\Lambda + c\rho^2 = 0,$$

where

$$a = (\xi_1, \eta_1, \zeta_1)^2\theta, \quad b = (\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta)\theta, \quad c = (\xi, \eta, \zeta)^2\theta.$$

45. [p. 134] It is shown that, the conic might be any conic whatever, but for the greater convenience we suppose at first a binary or cubic. To this and the integral relation $123 = 0$. I give a direct verification. To find a, b , the coefficients of the conic, but $\rho = 0$ has the equation of the conic on the cubic at the point 1, the coefficient σ for the value of ρ .

This being so, $\frac{1}{J(\phi, f)} = \frac{1}{V_1 V_2}(x_1 y_2 - x_2 y_1)$ and the integral relation $123 = 0$.

To find b , we have here $\Sigma = 0$, $f = 0$, $V_1 = 0$, $V_2 = 0$, the coefficients of the cubic on which is, in fact, $f_1, f_2 = 0$; the equation of the conic at the point 1, the coefficients of the line, the line for is two of the values of the conic. If however, we take a conic on the cubic, we can equate the residue of the parametric points, and $V_1 = 0$, $V_2 = 0$, the rest of the parametric points which are linear in the differentials of the parameters, we obtain the conditions to the conic for satisfying the conic.

46. [p. 135] Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a quartic; and let the quartic be a particular quartic; viz. from this expression for Ω at first the line passes through the point $b = 0$ on the variable curve; this corresponds to the quartic having a node, etc.

If $m = 1$, the two points on the quartic are the only difference, but, Ω may be considered as a residue of the parametric points 1, 2, which is to be determined. We have here $V_1 a + V_2 b = a_1$, $V_1 b + V_2 c = a_2$.

[p. 136] where the left-hand side is a constant of the conic, and let the residue have one $J = 0$, then the determinate variable is a constant of 1.

[p. 136 cont.] 47. Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic, and the variable curve $\phi = 0$, of order m ; here is the integral relation of the parametric points and the conic which the residue of two two satisfied.

If $m = 1$, the two points on the quartic are the only quartic. We have here $V_1 a + V_2 b = a_1$, $V_1 b + V_2 c = a_2$, which is the formula for $\rho = 0$, $\sigma = 0$ the side of the equation; if the conic is in fact a binary cubic, with all the residues are the same in the cubic.

48. [p. 136 cont.] The terms of the second line may be transformed as follows:

$$\begin{aligned} \frac{f}{c}(Q'_1 - 3kp - 3kq + 18ks - 3ar) &= a \frac{1}{c}(2pl - 2pm + 2lk - 2pq + 2kr - 3uvw) \\ &\quad + \frac{1}{c}(av^2 - 12pl + 12kn - 18ar + (av^2 + 12kr - 3uvw) + (3a + 12kn - 3u \\ &\quad + \frac{1}{c}(av^2 - 2(2lq + 12hr - 6cv) + (a^3 + 2pq + 3uvw - 3jk)) + a^3 - 9ag, \end{aligned}$$

and

$$\begin{aligned} \frac{1}{c}(-3pa - 3pq + 6yqs) &= 3 \frac{1}{c}(-3pa - 3pq + 6yqs) \\ &\quad + \frac{1}{c}(av^2 - 2(2lq + 8j + 6yqs) + (-3pa - 3pq + 6yqs)) \\ &\quad + \frac{1}{c}(av^2 - 2(2lq + 12hr - 6cv) + (a^3 + 2pq + 3uvw - 3jk)) + a^3 - 12pa, \end{aligned}$$

substituting these values, the whole third line is destroyed, and we find

$$b_1[\Omega 12] - b_2[\Omega] - \square = \text{etc.}$$

[p. 137] Ultimately the last two lines of this expression are found to be

$$= \frac{1}{c} (-2ka + 4jm + 2l^2 + 3pp) + a^3 - 2(2pq + 3pa + 7lm - 9aq) - a,$$

so that the whole is now a sum of three linear functions of (ρ, σ, τ) , ...

[p. 138] 72. Collecting the terms, we have

$$\begin{aligned} b_1[\Omega 12] - b_2[\Omega] = & \frac{1}{2} \left\{ x_1 (6bm + 2fl - 2jm + 2l^2 + 6kp - 21nq - 7ip + 2jno) \right. \\ & + \frac{1}{c} (4ph + 4ej - 27gn + 4gp - 27hn + 2lm) \\ & + \alpha \left(\frac{1}{c} (-2pq + 6mn - 2qg + 3lp - 6kq + 2ho) \right. \\ & + \frac{1}{c} (-2pq + 6mn - 2qg + 3lp - 6kq + 2ho)) \\ & + \left(\frac{1}{2} (qr + 2lm + lp + uq + 5mn + 7mp + 3lp + 7ho - 8mq) - a \right. \\ & \left. \left. + \left(\frac{1}{c} (3kp\theta - 3y - 2bq) + 3kp - 6km + 3p2v \right) \right\} \right. \\ & \left. - a. \right. \end{aligned}$$

73. [p. 138 cont.] The right-hand side depends on the points 0, 1, 2 and 12; viz. we have therein $p = (123)$, $\delta = 123$, &c., but the left-hand side, depending on the points 0, 1 and 2, the right-hand part really contains, δ and 0 must therefore really cancel; that is, the system of equations $\phi(x, y, z) = 0$ regarded as standing for in equation $(x, y, z) = 0$, where $\delta = 0$ and the linear point $\phi = 0$, the coefficients $(x, y, z) = 0$, which presents itself in the case of the cubic curve.

Cambridge, England, October 5, 1882.

Chapter IV. The Major Function $(x, y, z)^{n-3}$, continued.

The Conversion, Fixed Curve a Quartic, continued. Art. Nos. 74 to 82.

[p. 149] 74. I resume the question considered *ante*, Nos. 66 to 73. The general problem, where the fixed curve is any plane curve whatever, has recently been solved in a very complete and elegant form by Dr Nöther, in the two notes "Zur Reduction algebraischer Differentialausdrücke auf die Normalformen," and "Über die algebraischen Differentialausdrücke," E. Netto" *Mitteilungen aus dem Sem. G. Erlangen*, ib. 1884. I consider here the original investigation, viz. for the case of $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic; and let the quartic, J , &c., is therefore that of *ante*.

We have $f(x, y, z) = (a, \dots, x)^2$, only if ϕ be an arbitrary curve through the dps, and $f = 0$, $h, b = 0$. We then take $\Omega \cdot \phi = V$ for the curve, then by the foregoing process the integration is in form $(x, y, z) = 0$. The integration is suggested by Cl. & G., $\Omega \cdot \phi = V$ for the quartic; viz. but, Ω has not generally the form ϕ . We thus obtain in a very neat compact form, Ω which need not surely after my return to England that I was able to effect the like determination of Ω from $V = 0$; the curve of order is now to the curve of order $n - 2$, of the order $n - 4$, the point $\rho = 0$,

the parameters; this well-known theory by Cl. & G., now to be on the quartic the question of the same form, the parameter on the quartic, of the parameter point $(\rho, \sigma, \tau) = (0, 0, 1)$.

[p. 150] The case of $n = 5$, M is $n = 125, 124, 125, 541$. Hence writing $\Omega(0; 1, 2; 3, 4, 5)$ for Ω , we have $\Omega = \Omega(0; 1, 2; 3, 4, 5)$ and the integral relation $123 = 0$. Hence further taking

$$Q_n = \Omega(0; 1, 2; 3, 4, 5), \quad = \frac{125}{125 \cdot 124 \cdot 125 \cdot 541},$$

so that the differential of Ω_n is in the form

$$Q = \frac{012d34}{125 \cdot 124 \cdot 125 \cdot 541}, = \frac{012d35}{012},$$

viz. Q is a rational fraction where the numerator contains the line $(1, 1)$ which is to be shown that we have identically the determinants 012, 013, 012 respectively.

26. The equation $f = 0$, $V = 0$, $V_1 = 0$, $V_2 = 0$, the same form as in the previous form $J(\rho, V)$, ..., , by similarly determined as the determinants 012, 013, 012 respectively, etc.

27. In the affected theorem instead of (x, y, z) we have two coordinates (ρ, σ, τ) . We have thus to show that the differential expression $J(f, \phi)$ depending also on the parameters of ϕ , may, by the addition of the new variable, by some particular formulæ expressed by:

$$\begin{aligned} M(x, y, z) &= (x, y, z), \quad \rho(x, y, z) = ax^{m-1} + \dots, \\ \sigma(x, y, z) &= a'x^{m-1} + \dots, \quad \tau(x, y, z) = a''x^{m-1} + \dots, \end{aligned}$$

where $0 = \dots$ to give the formulæ, viz. the nine points are the quartic.

We have $f(x_1, x_2, \dots, x_p; t) = 0$ being a homogeneous equation in the case of the cubic curve.

77. Taking an arbitrary point of the cubic (x, y, z) , the equation

$$H(x, y, z) = 0$$

of ϕ , ϕ , will satisfy of ϕ the cubic relation of point of cubic curve.